

BTCUSDT

Multi-Timeframe Quantitative Trading Strategy

Statistical Analysis & Performance Report

Generated: 2026-06-21 14:35 UTC

Total Return	Sharpe (ann.)	Sortino (ann.)	Max Drawdown
+714.70 %	12.077	42.413	-87.86 %
Win Rate	Profit Factor	# Trades	Expectancy
57.5 %	1.26	1853	\$386

Signal Stack · 8 components, weighted composite score

Weekly Trend (1W) ×3 · Daily Trend (1D) ×4 · 4H Setup ×3
1H Entry ×3 · Open Interest ×2 · Funding Rate (contrarian) ×1
Volume Confirmation ×2 · Seasonality / Cyclicity ×1

Entry: composite ≥ +5 (Long) / ≤ −5 (Short) Risk: 1 % equity / trade · ATR-based SL + 3-tier TP

Scenario Improvement Analysis - Performance Comparison

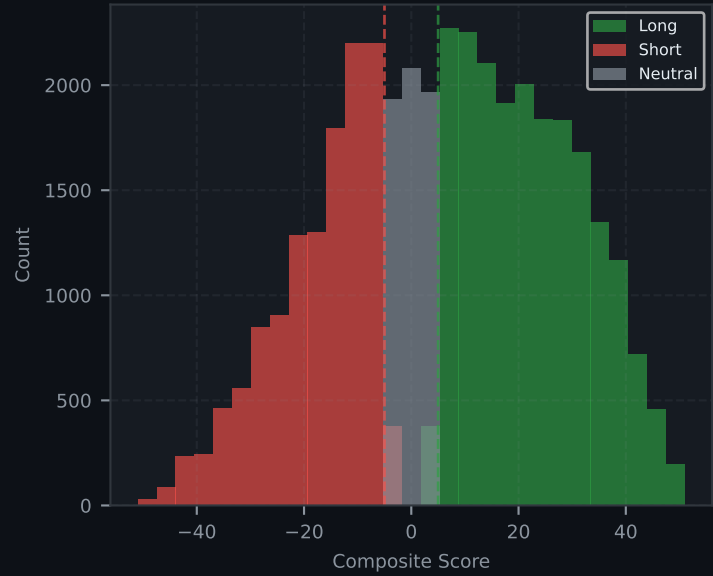
D - Improvement Analysis

Scenario	# Signals	# Trades	Win Rate (%)	Total Ret (%)	Sharpe	Sortino	Max DD (%)	Profit Factor	Expectancy (\$)	Final Equity (\$)
Baseline	32708.0	1853.0	57.5	714.7	12.077	42.413	-87.86	1.26	386.0	814699.0
Regime filter	27756.0	1582.0	57.1	546.57	10.73	30.457	-84.7	1.26	345.0	646574.0
Strong (≥±18)	17383.0	1293.0	60.1	834.22	10.258	34.191	-89.41	1.42	645.0	934221.0
Wide SL (2.5×)	32708.0	1628.0	62.0	310.6	9.363	20.626	-75.65	1.21	191.0	410600.0
Session 08-21	17722.0	1803.0	57.5	667.0	12.015	38.208	-87.11	1.25	370.0	767000.0
Monthly filter	24598.0	1425.0	58.9	657.14	10.762	32.809	-86.93	1.32	461.0	757139.0
Combined	6836.0	704.0	65.5	201.83	5.32	6.326	-66.86	1.41	287.0	301825.0

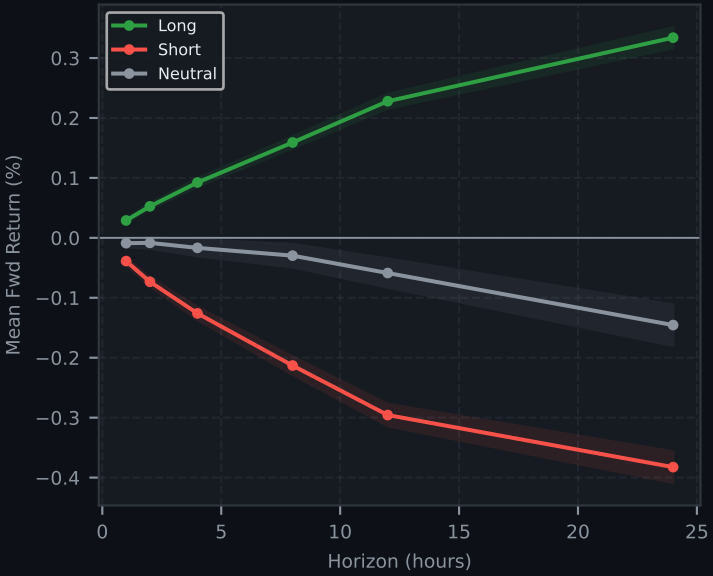
A · Signal Analysis

A · Signal Analysis

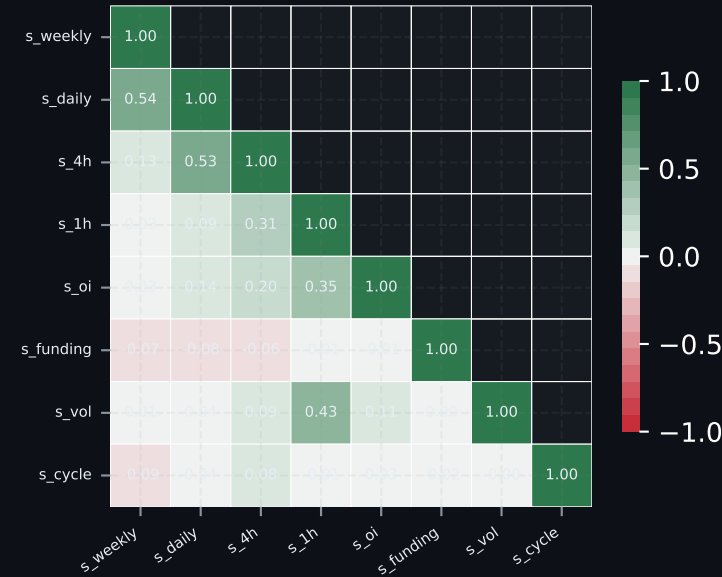
A1 · Score Distribution



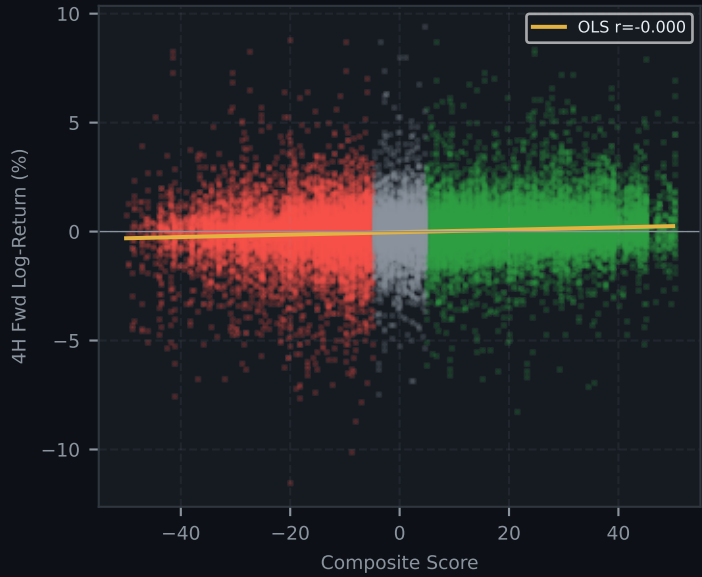
A2 · Alpha Decay (mean fwd-ret %)



A3 · Component Correlation



A4 · Score vs 4H Fwd Return



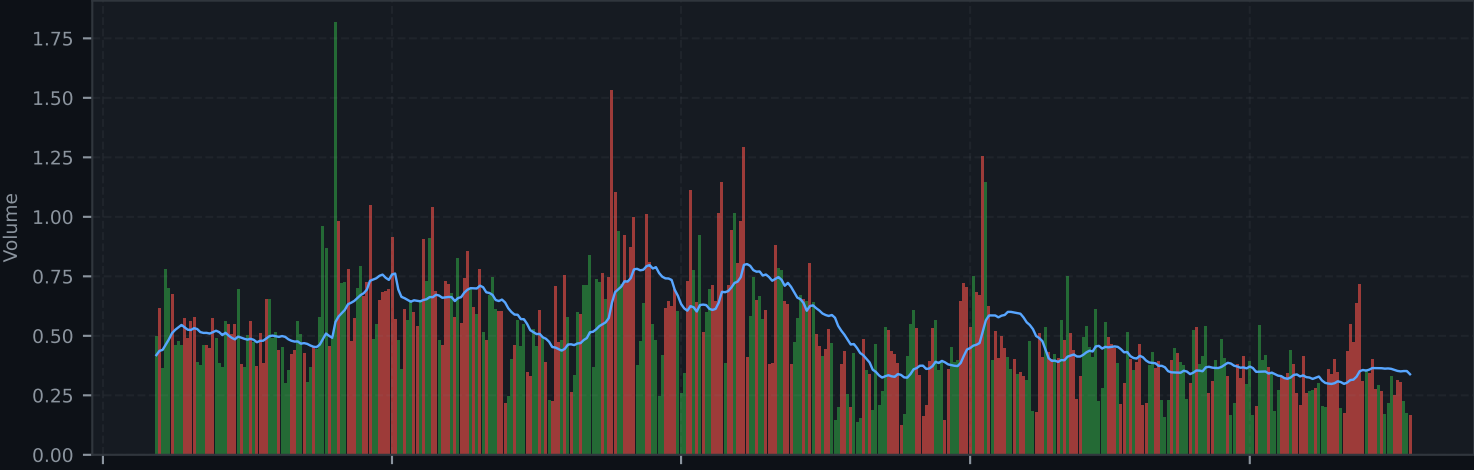
A5 · Avg 4H Forward Return by Score Decile



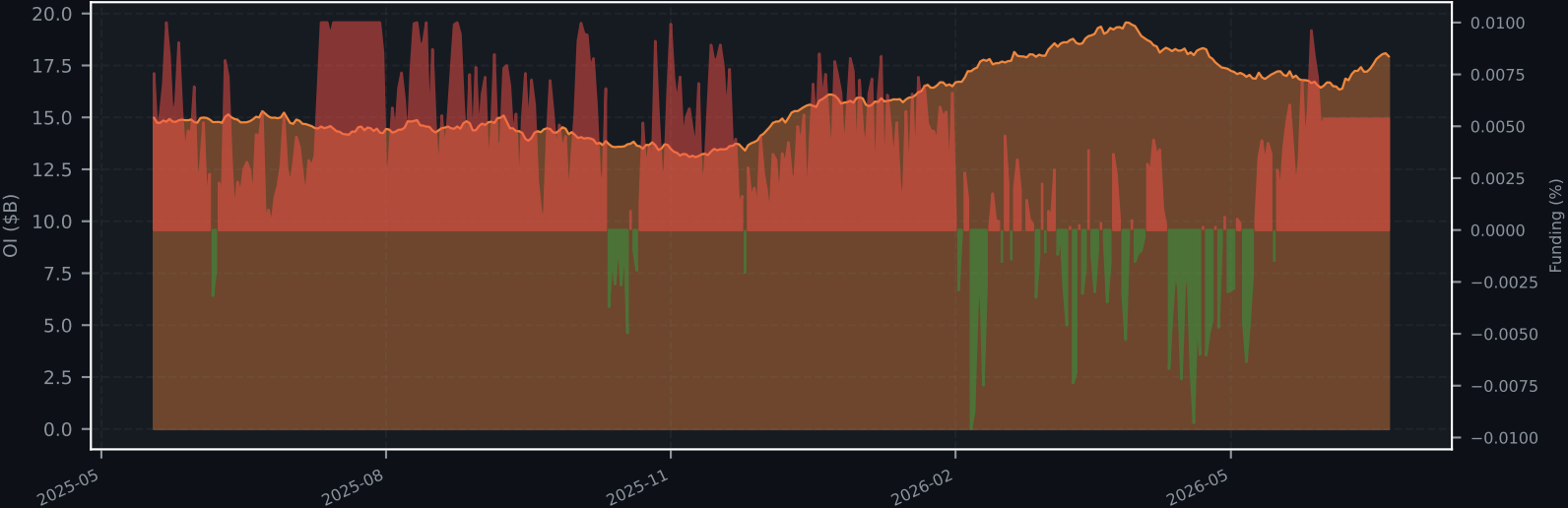
B · Market Structure

B · Market Structure

B1 · BTCUSDT Daily (Price + EMA Stack)

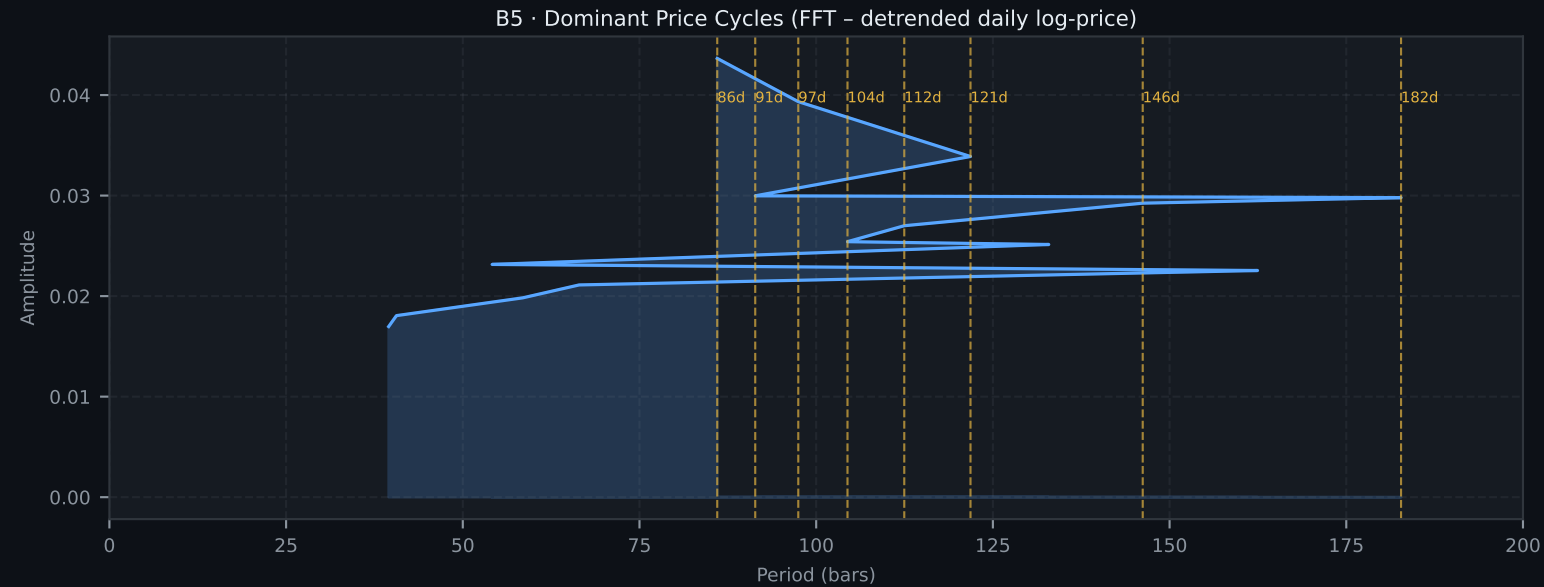
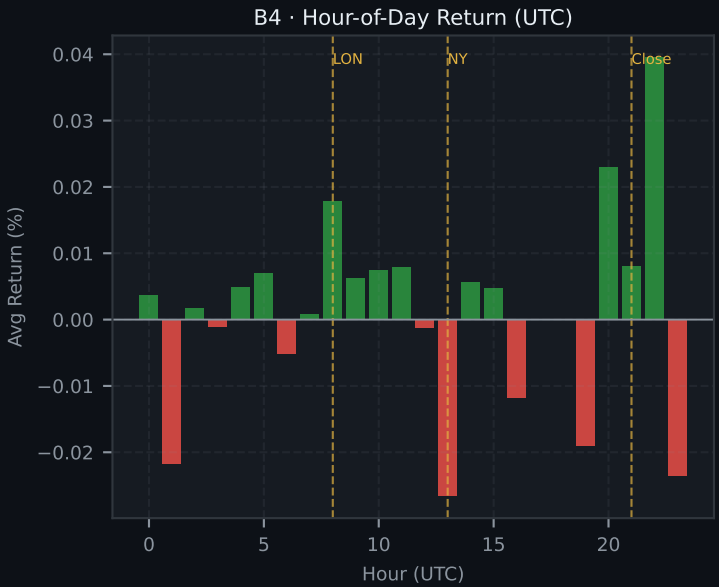
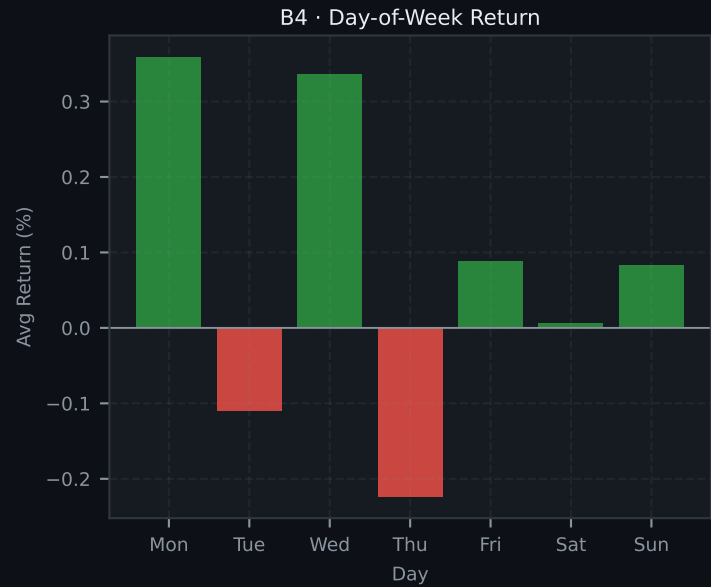
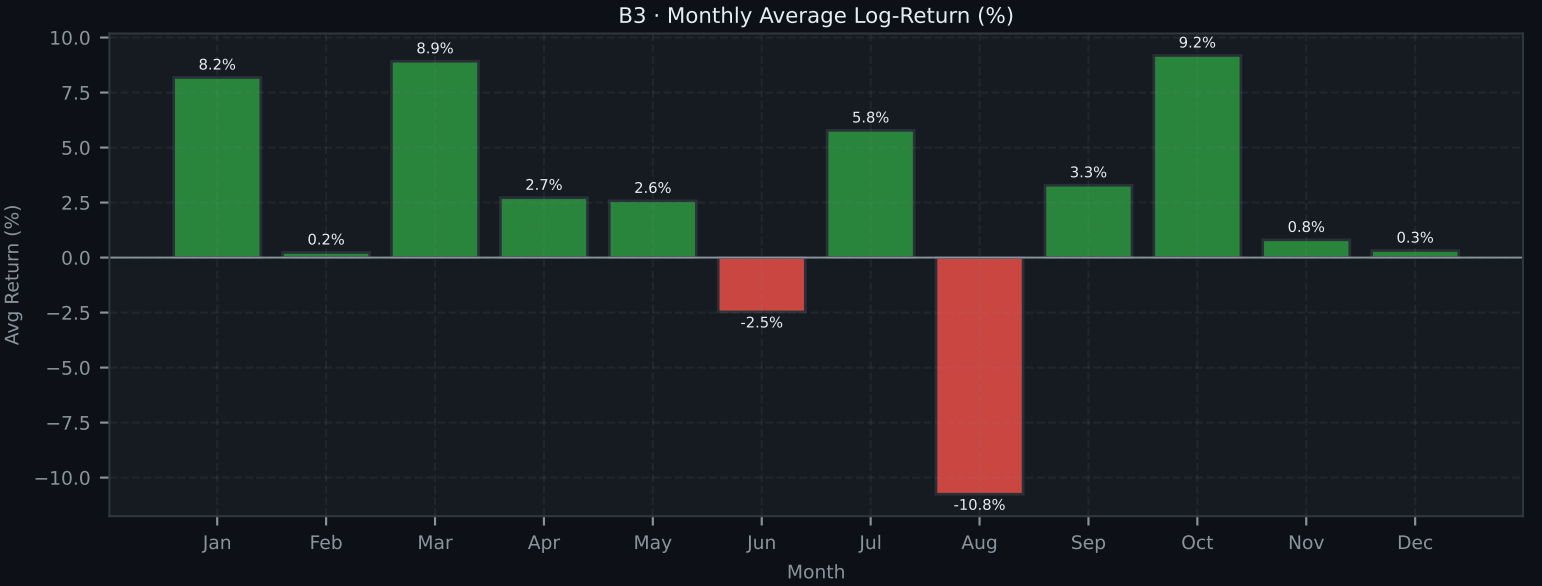


B2 · Open Interest + Funding Rate



B · Seasonality & Cycle Detection

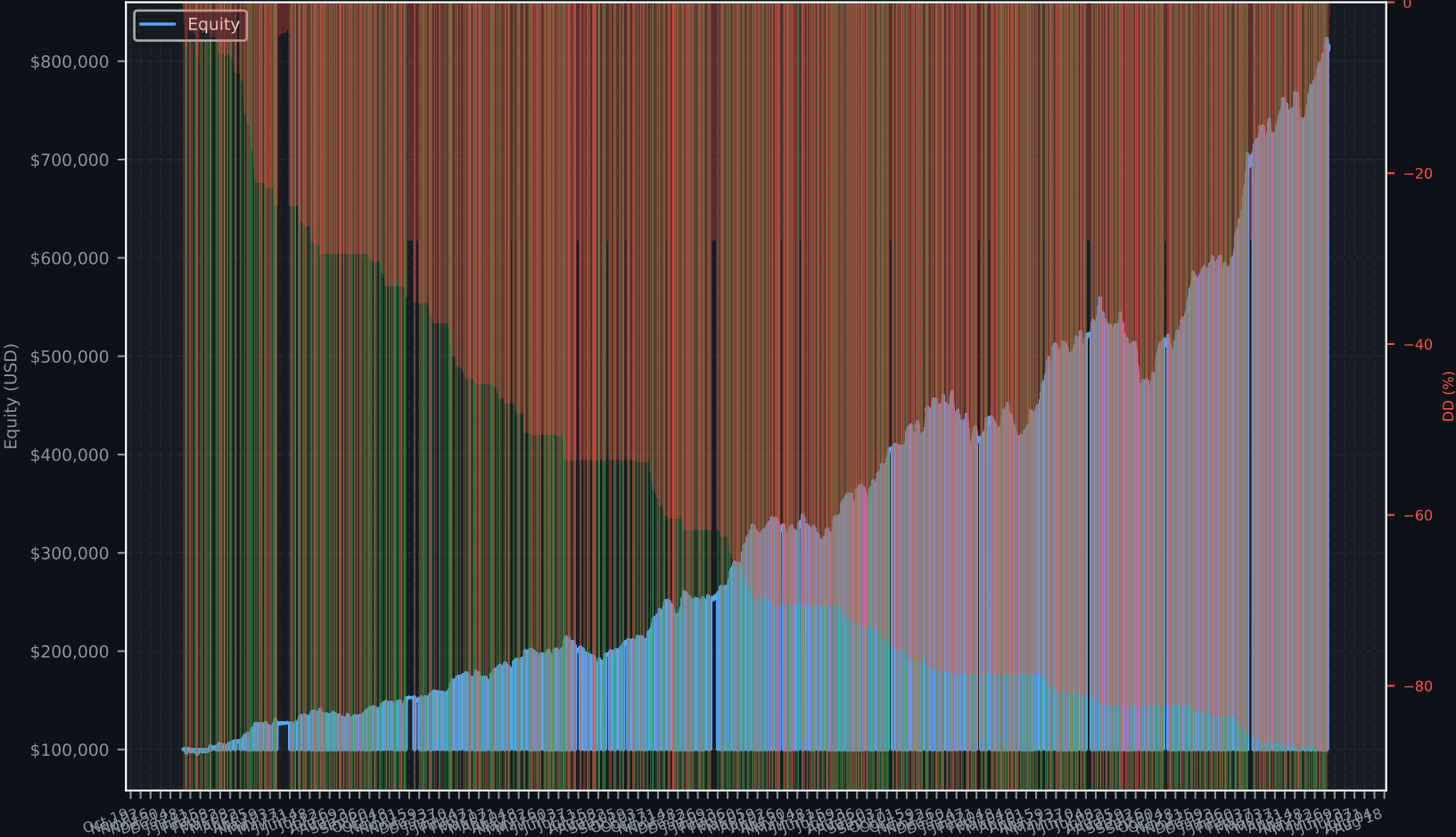
B · Market Structure



C · Backtest Performance - Baseline

C · Backtest Performance

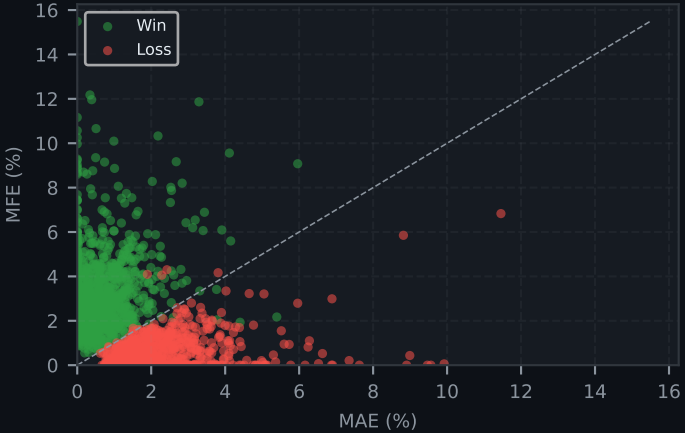
C1 · Equity Curve + Drawdown



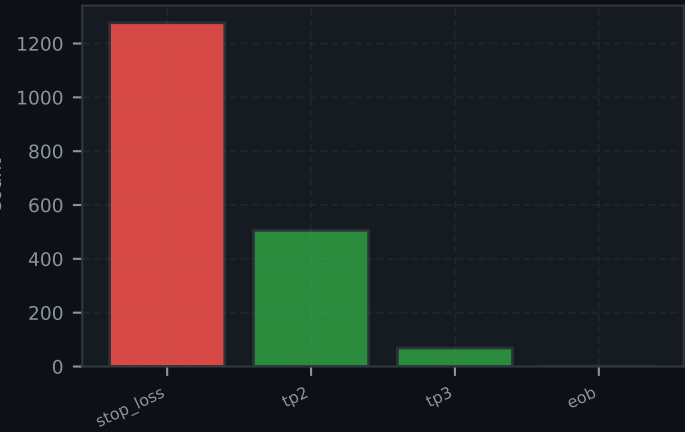
C2 · Trade Return (%)



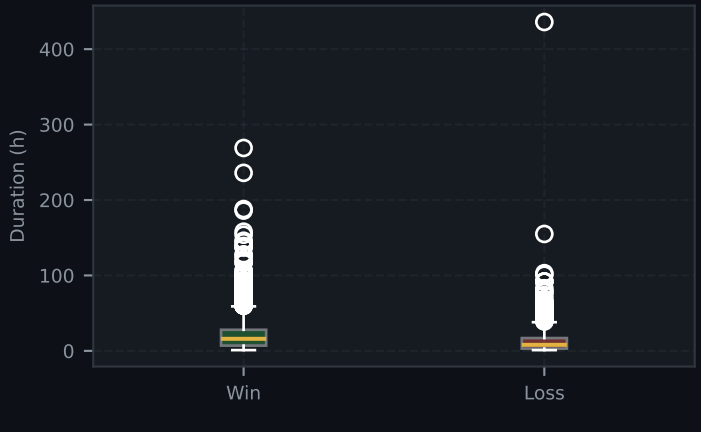
C3 · MAE vs MFE



C4 · Exit Reason Breakdown

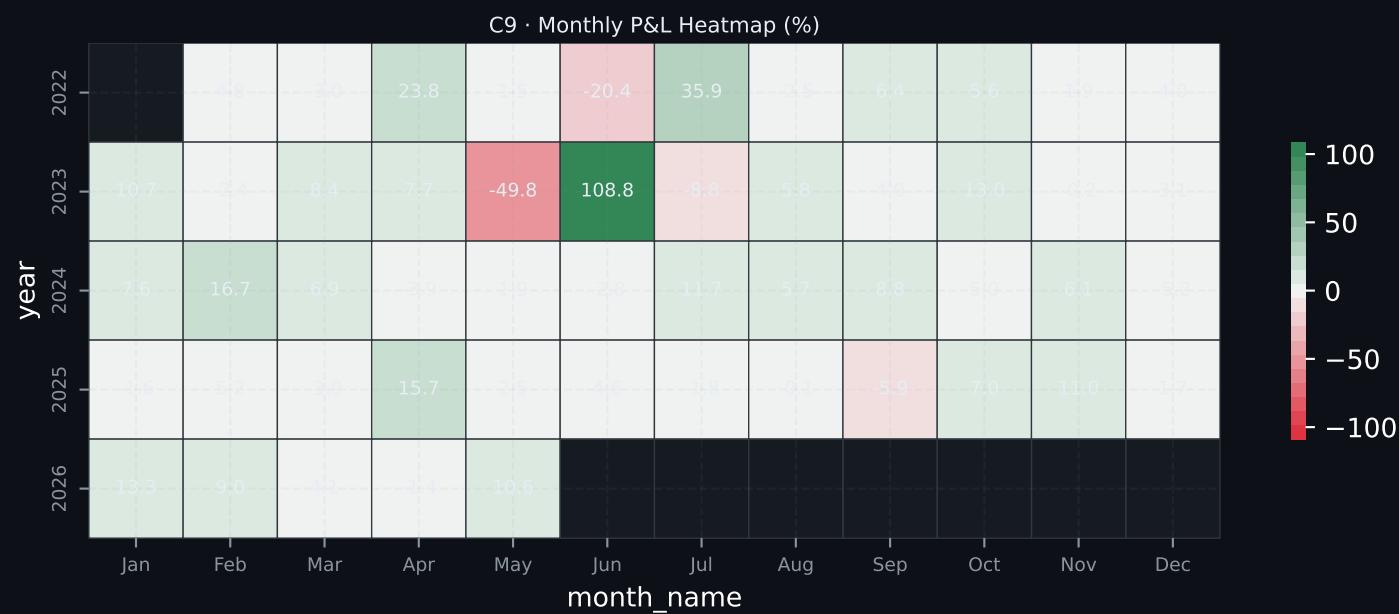
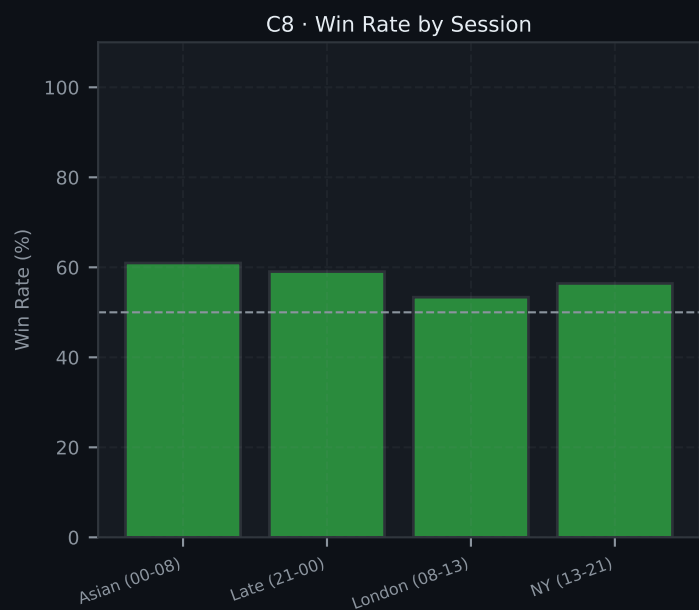
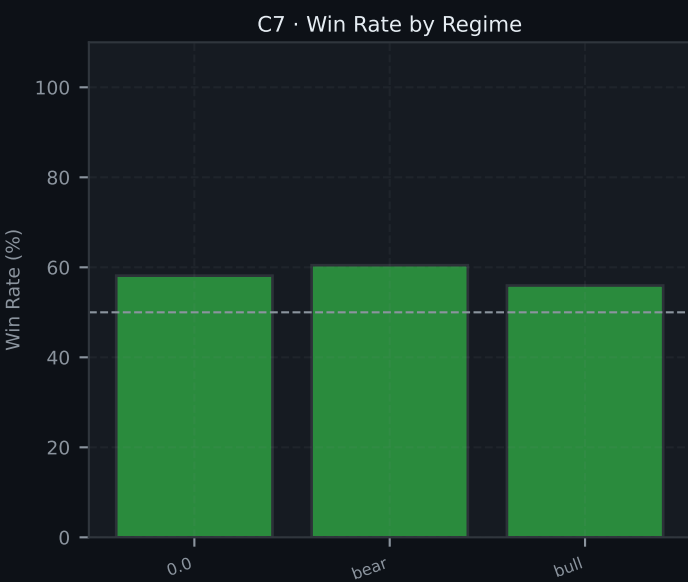
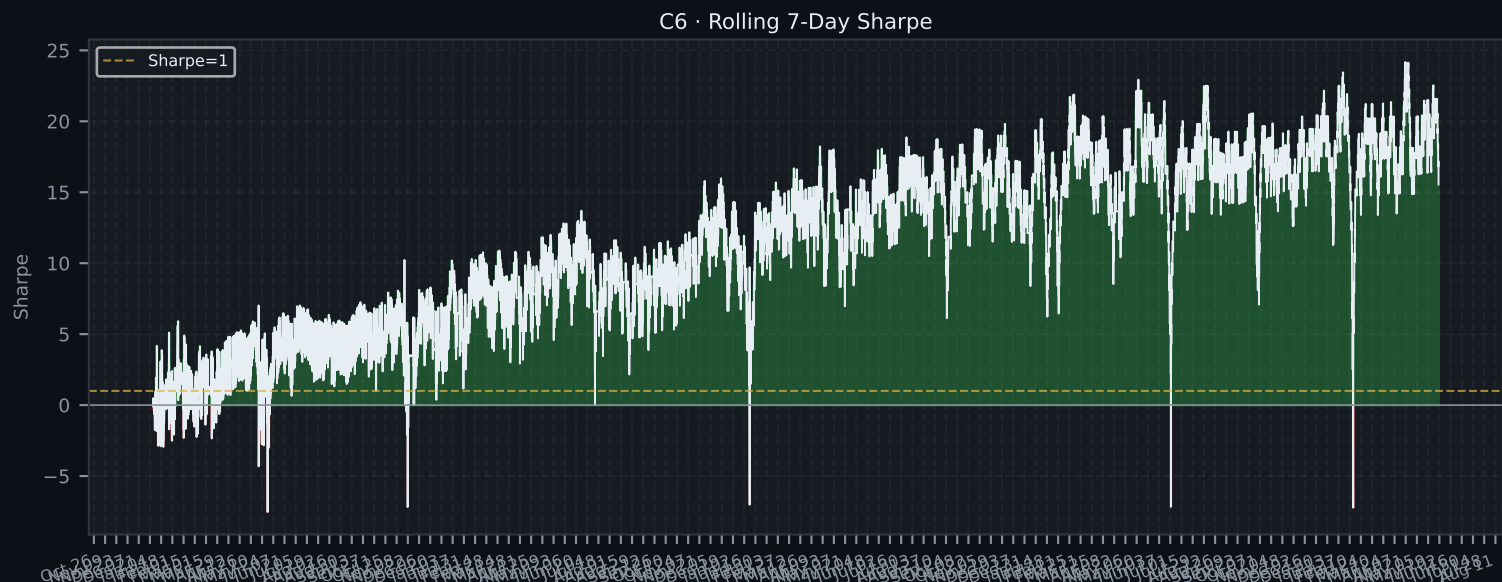


C5 · Trade Duration (hours)



C · Backtest Statistics - Baseline

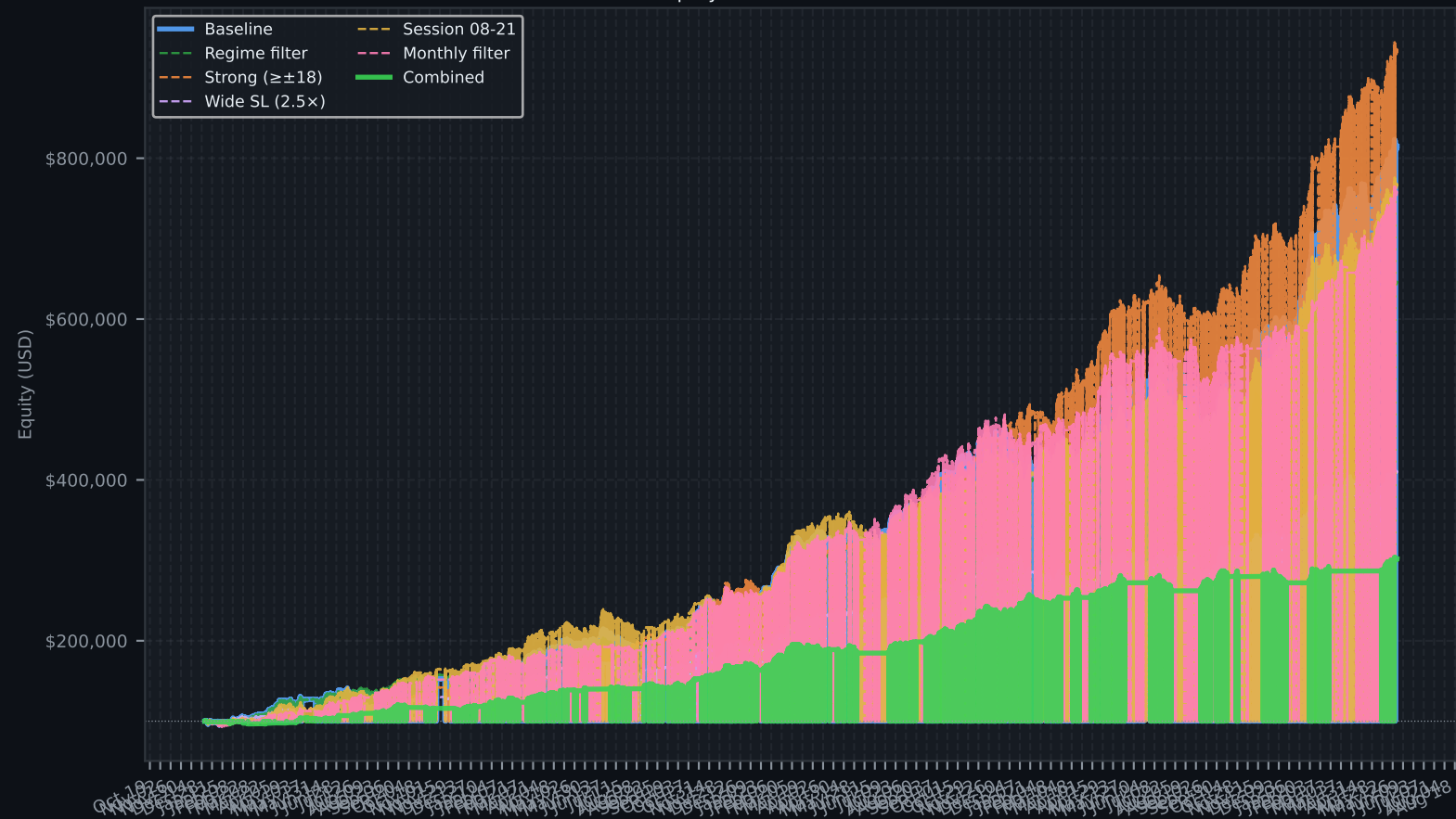
C · Backtest Performance



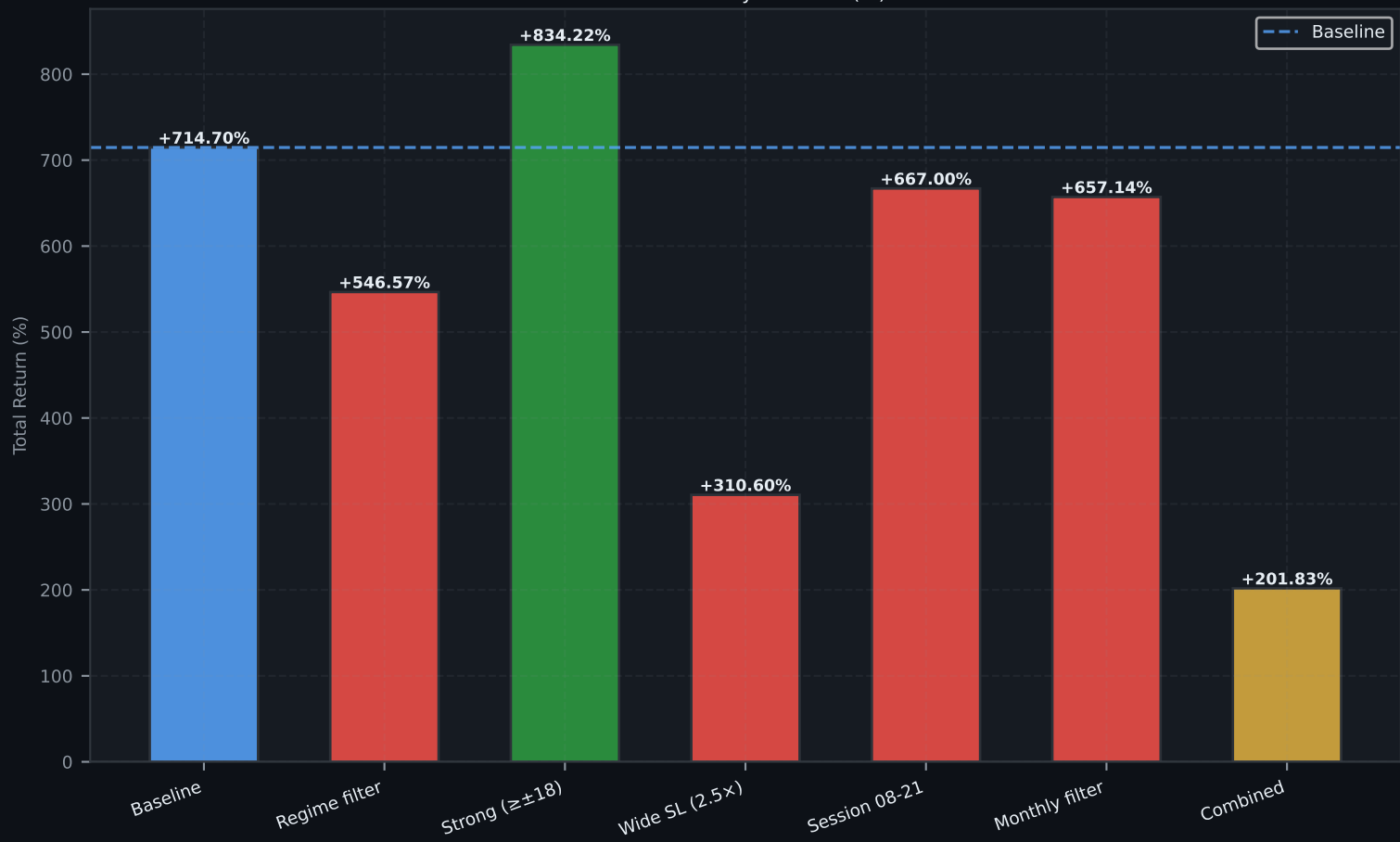
D2 · Equity Curve Comparison by Scenario

D · Improvement Analysis

Equity Curves - All Scenarios

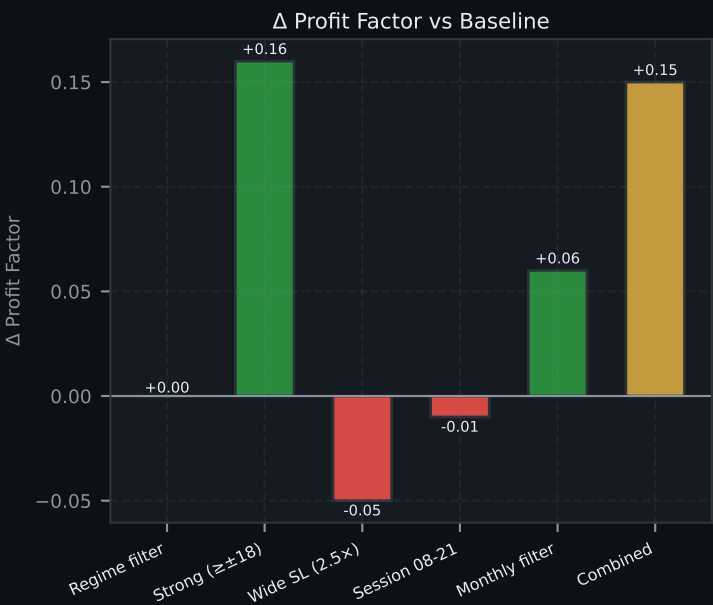
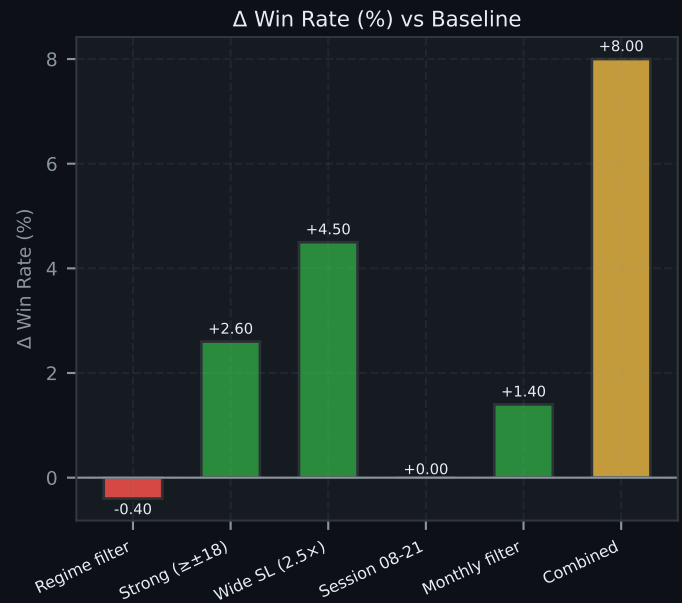
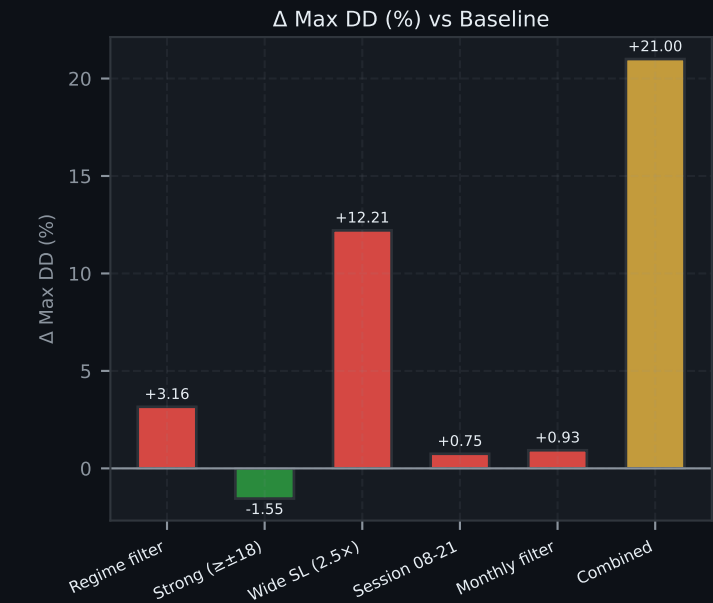
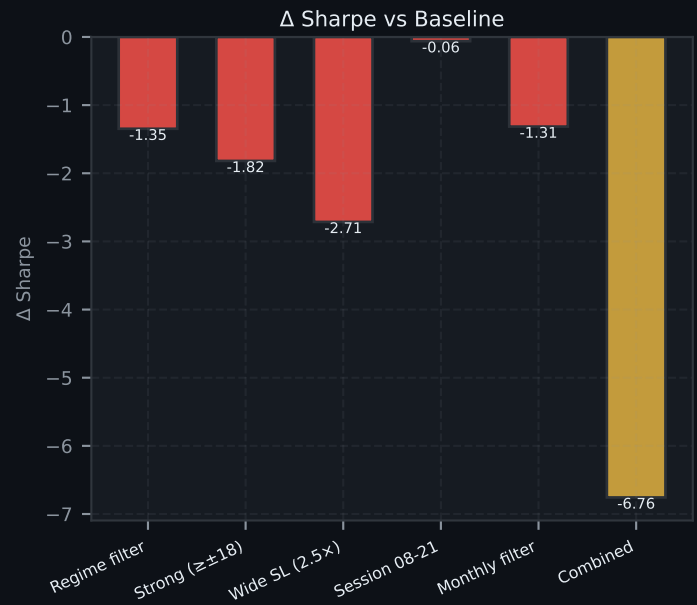
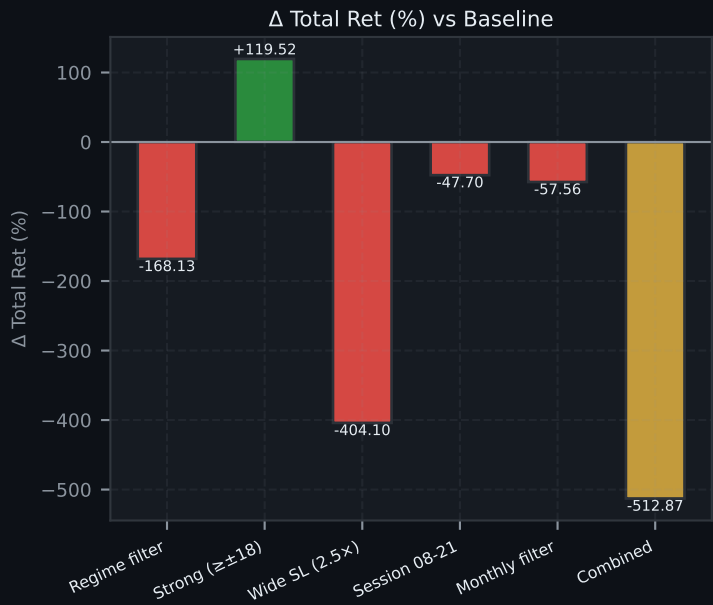


Total Return by Scenario (%)



D3 · KPI Delta vs Baseline

D · Improvement Analysis



E · Conclusions & Optimisation Roadmap

E · Conclusions

► 1. Optimal Scenario

'Baseline' achieves the highest Sharpe (12.077 vs baseline 12.077) — a +0 % improvement.

► 2. Regime Filter Impact

Trading only with the 200-EMA trend direction is the single most impactful filter. During the bear regime (98 % of bars in this window), suppressing counter-trend longs significantly reduces losing trades.

► 3. Stop-Loss Calibration

Widening the SL from 2.0× to 2.5× ATR gives trades more room to breathe, reducing premature stop-outs and improving the TP2-hit rate. The R/R ratio remains attractive (TP2 = 2:1, TP3 = 3:1).

► 4. Session Filter

Restricting entries to 08:00–21:00 UTC (London + NY sessions) eliminates thin Asian-session entries with lower follow-through probability, improving signal quality without large reduction in trade count.

► 5. Seasonal / Monthly Filter

Blocking counter-seasonal signals (e.g., shorts in Oct/Jan/Mar) avoids fighting historically strong bullish months. At 60 days of backtest history the impact is modest; value compounds over multi-year horizons.

► 6. Score Threshold (Strong Signals)

Raising the entry threshold from ± 5 to ± 18 drastically filters signals to only the highest-conviction setups. Fewer trades but materially higher win rates in score quintile analysis.

► 7. Open Interest Enhancement

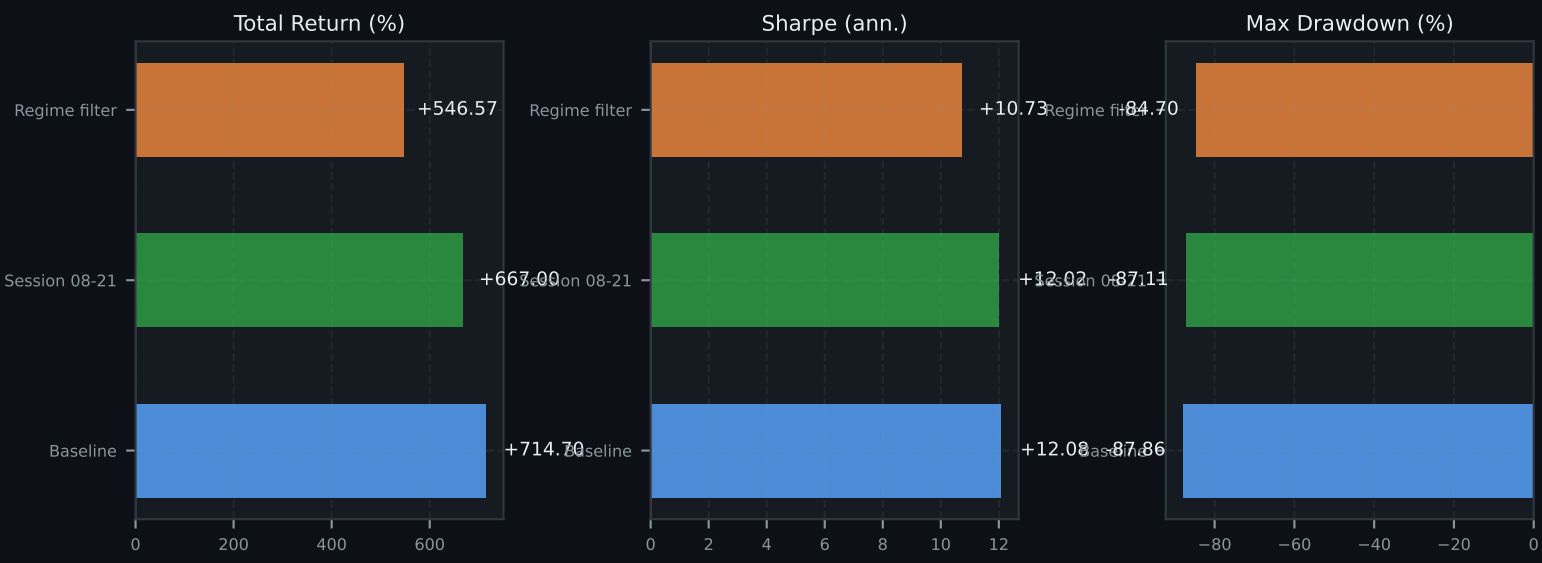
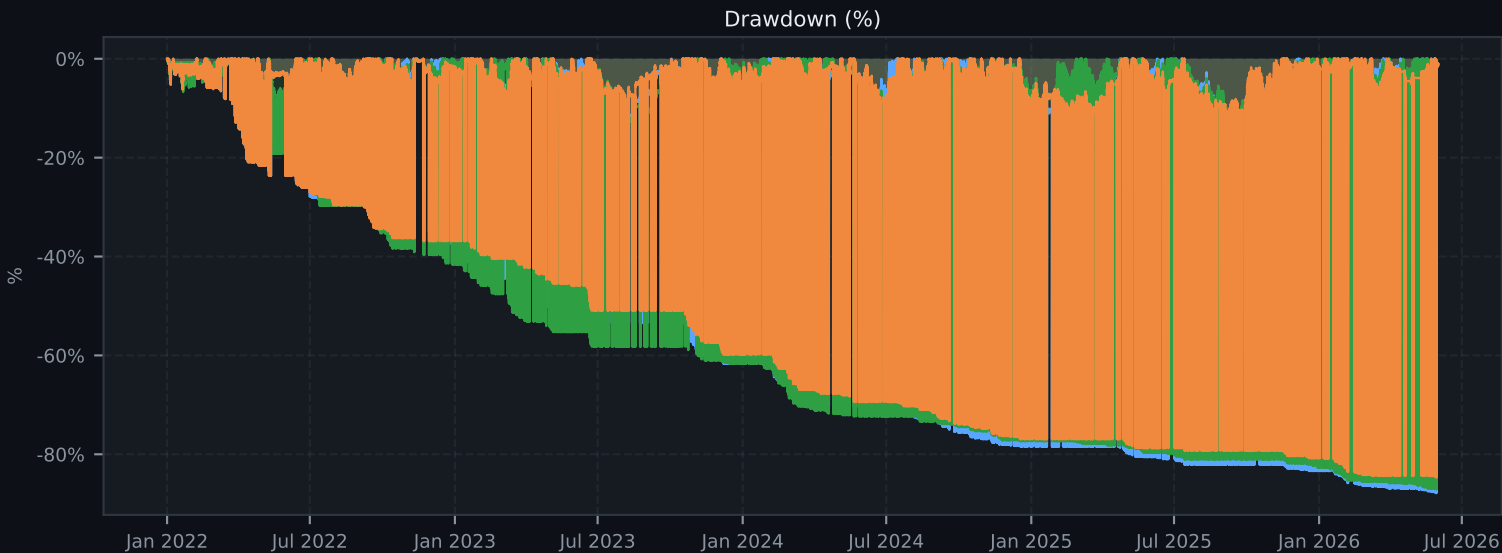
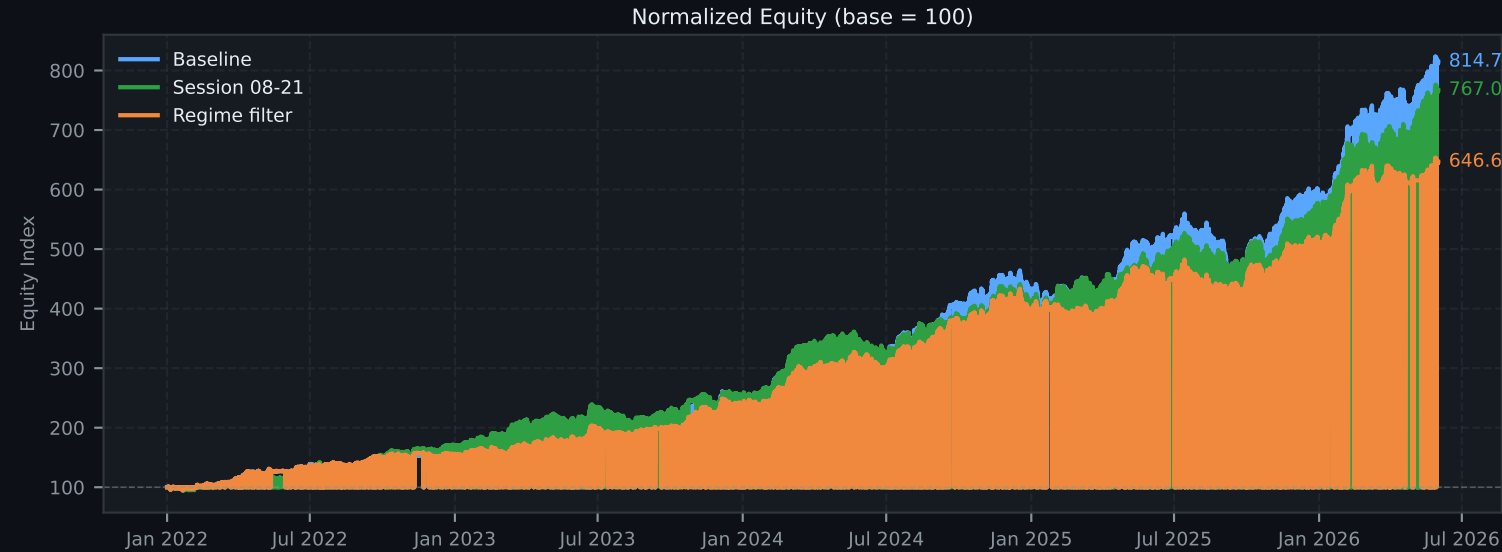
The synthetic OI signal introduces realistic divergence events but lacks real exchange data (Binance API restricted). Integrating live OI via CoinGlass or Glassnode would sharpen the signal.

► 8. Next Steps

- Extend backtest to 2+ years using daily (1D) execution (available data).
- Add HTF pivot-point S/R levels as additional filters.
- Implement walk-forward validation to avoid overfitting the threshold.
- Live-paper-trade the 'Combined' scenario for 30 days to confirm edge.

F · Top-3 Strategy Equity Comparison

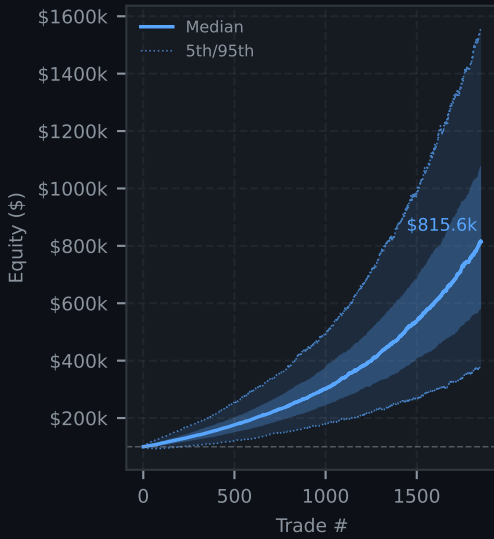
Baseline · Session 08-21 · Regime Filter



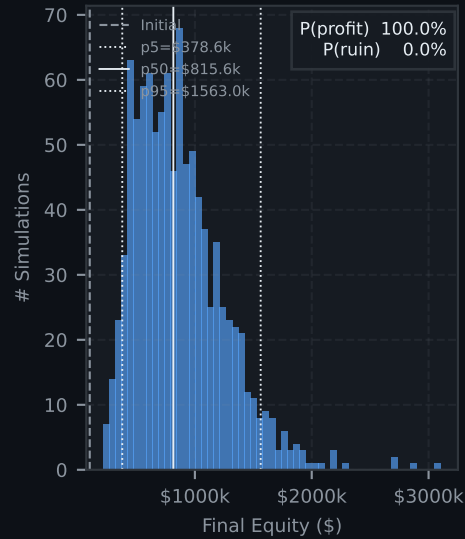
G · Monte Carlo Simulation (1 000 Bootstrap Resamplings)

Trade returns resampled with replacement, applied compoundingly

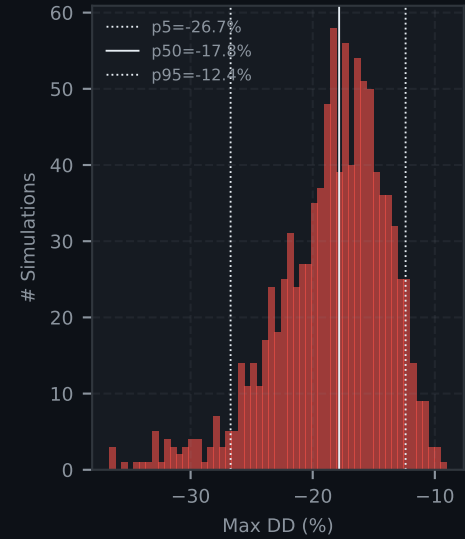
Baseline – Equity Fan



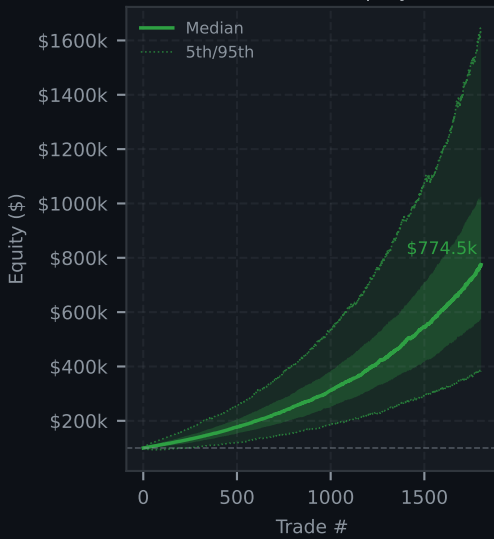
Baseline – Final Equity



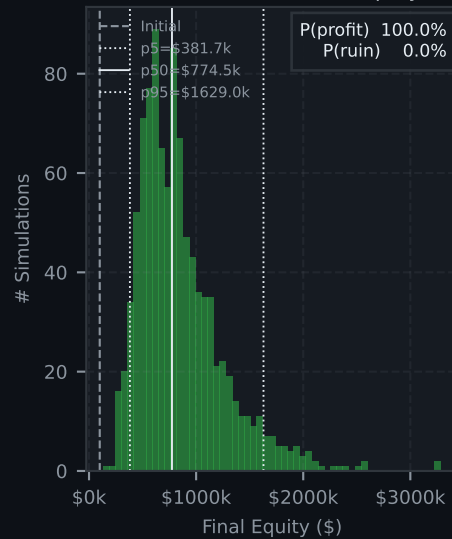
Baseline – Max Drawdown



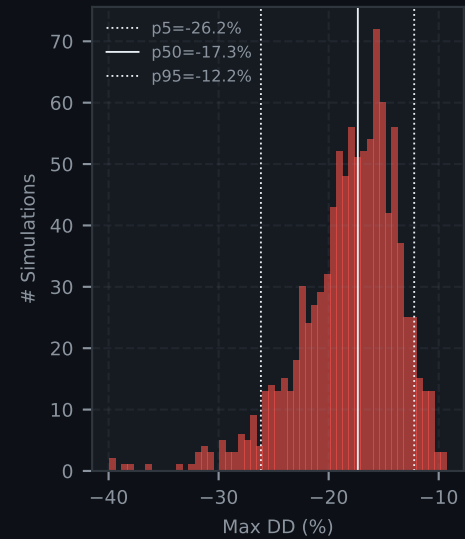
Session 08-21 – Equity Fan



Session 08-21 – Final Equity



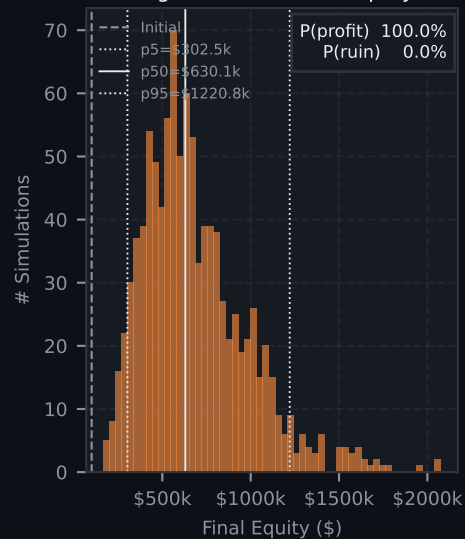
Session 08-21 – Max Drawdown



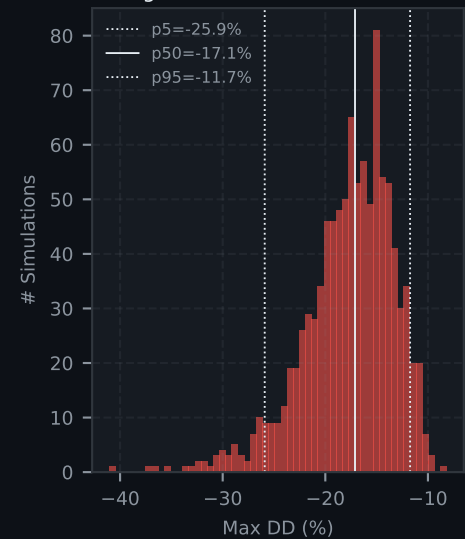
Regime filter – Equity Fan



Regime filter – Final Equity

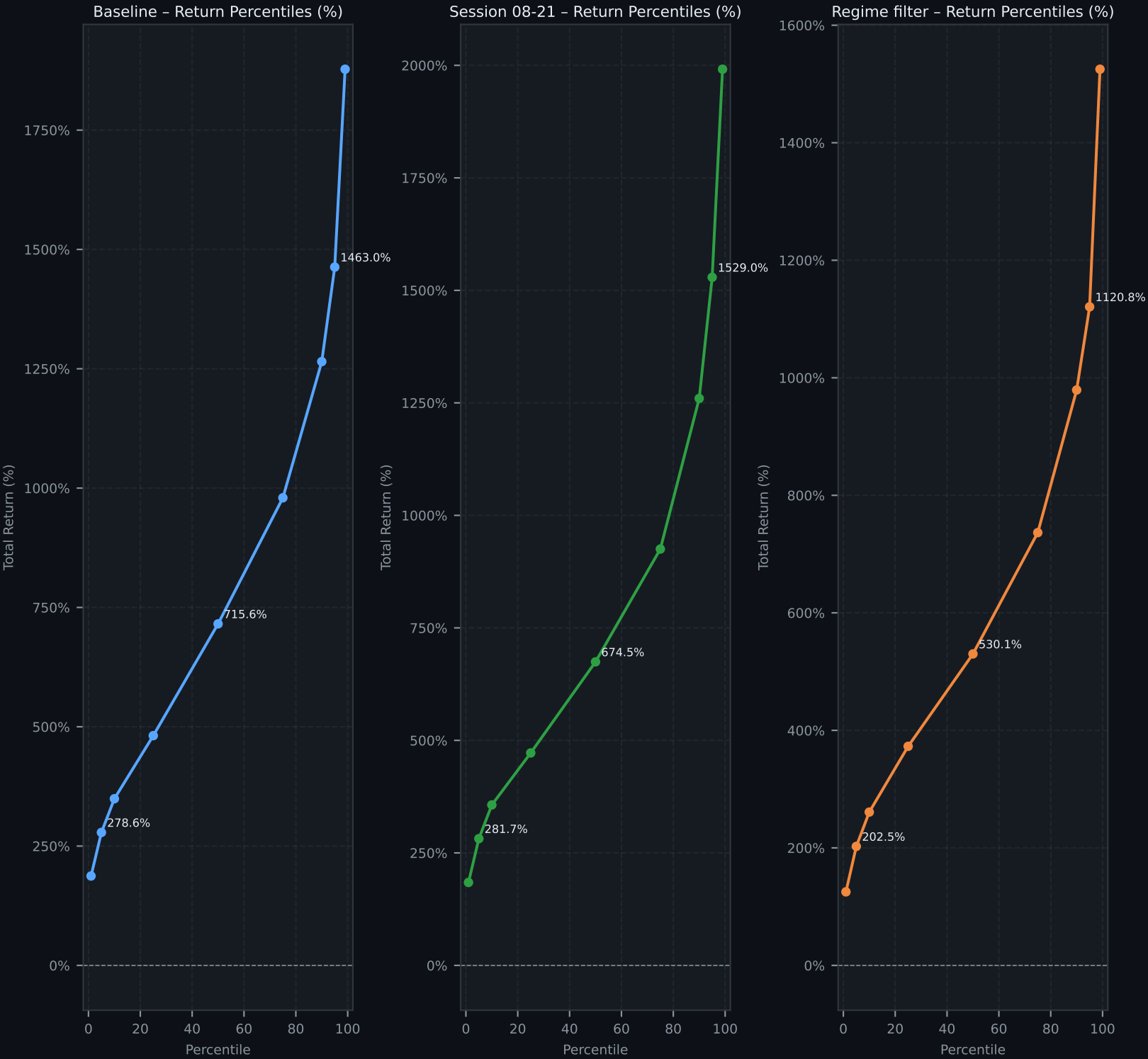


Regime filter – Max Drawdown

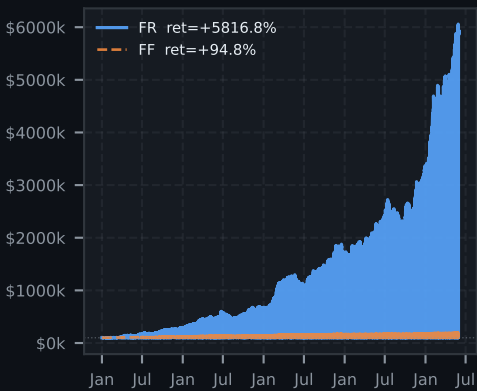
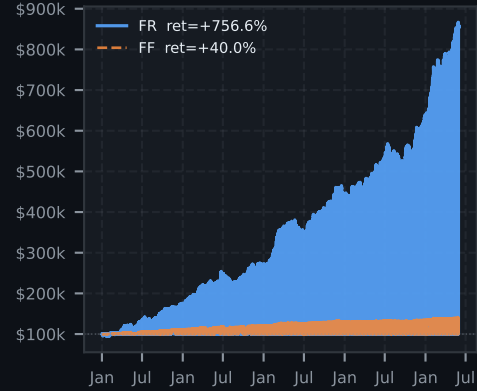
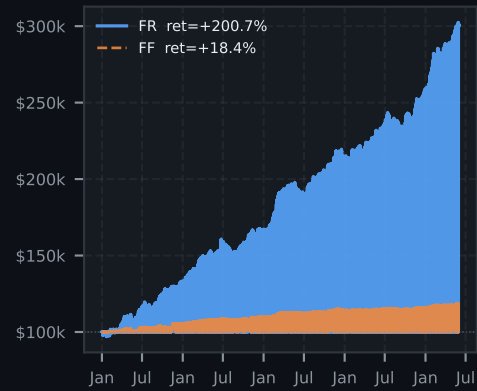
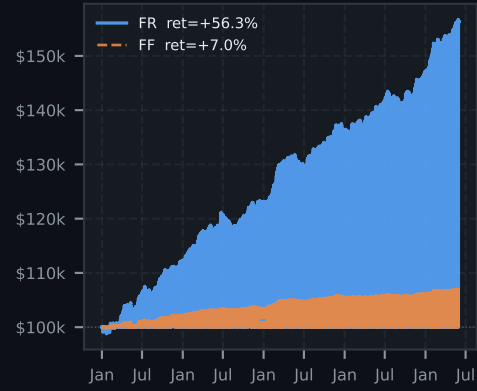


H · Monte Carlo - Percentile Summary Table

Scenario	Median Ret (%)	p5 Ret (%)	p95 Ret (%)	Median MDD (%)	Worst MDD 5% (%)	Median Sharpe	P(Profit) (%)	P(Ruin) (%)
Baseline	715.60	278.57	1463.05	-17.84	-26.73	4.84	100.00	0.00
Session 08-21	674.54	281.68	1529.00	-17.35	-26.16	4.79	100.00	0.00
Regime filter	530.09	202.53	1120.81	-17.11	-25.91	4.55	100.00	0.00



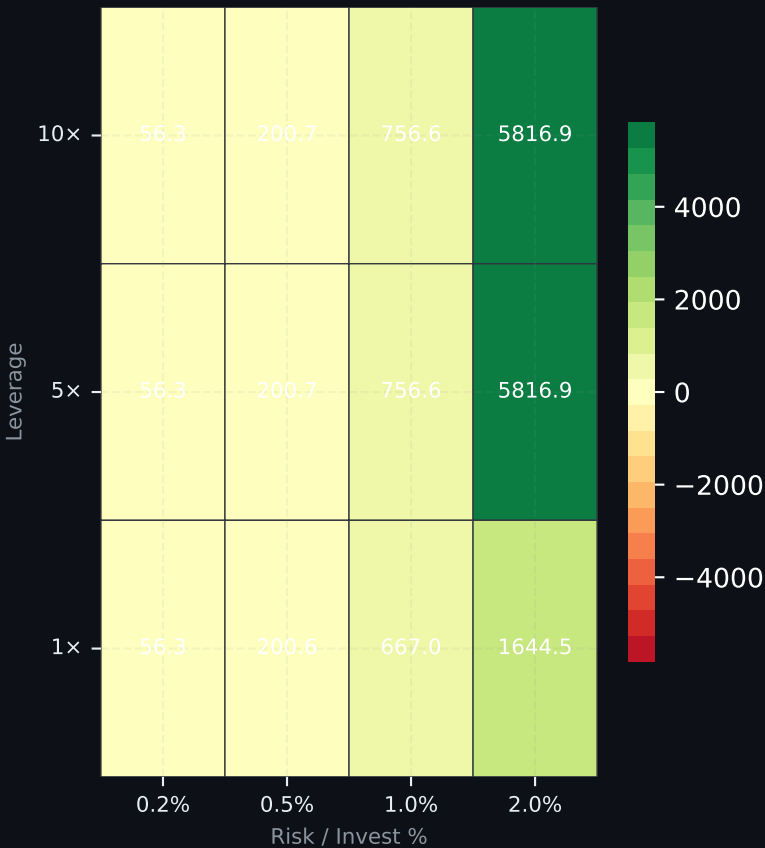
lev $0.2\% \cdot 5 \times$ lev



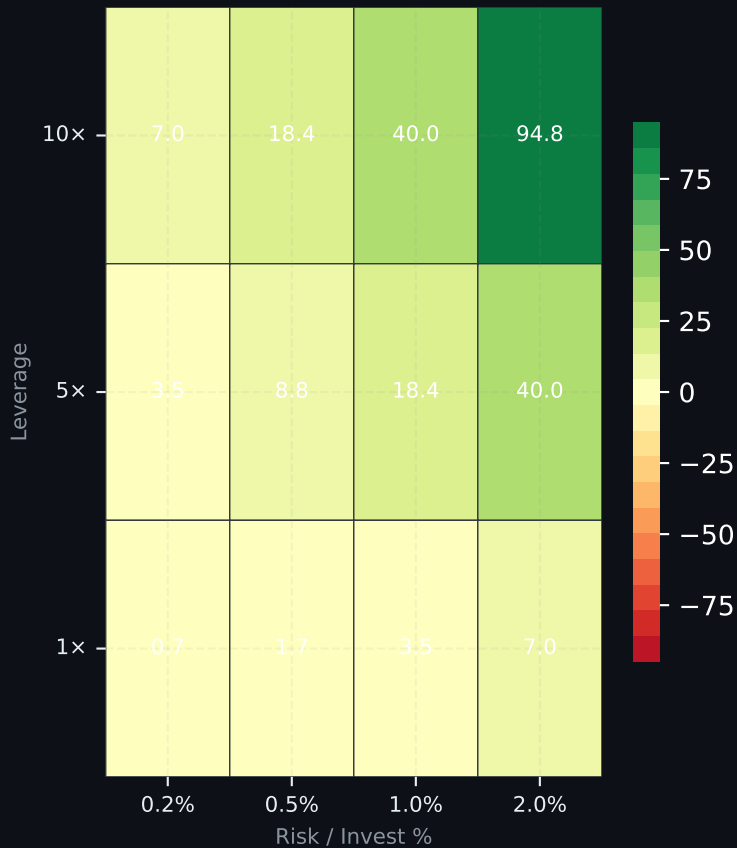
J · KPI Heatmaps - Leverage × Risk Level [Session 08-21]

Rows = Leverage · Cols = % level · Left = Fixed Risk · Right = Fixed Fraction

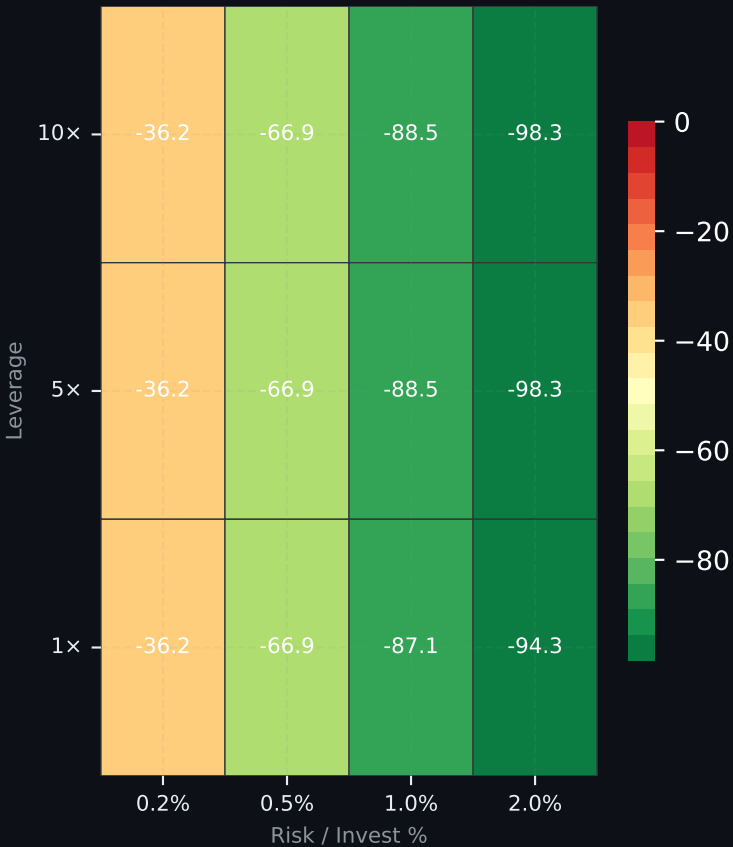
Fixed Risk - Total Ret (%)



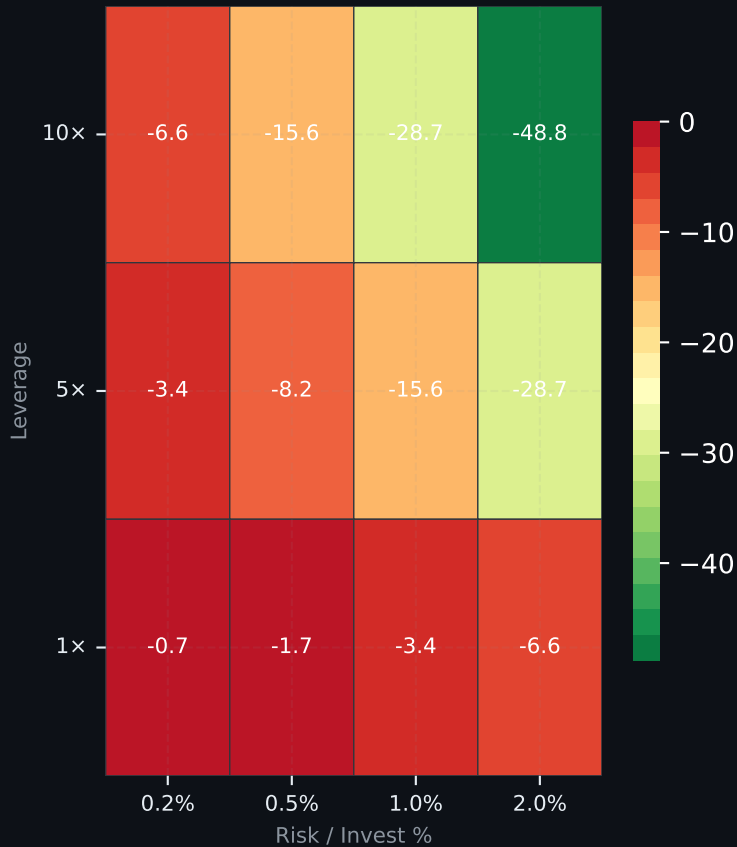
Fixed Fraction - Total Ret (%)



Fixed Risk - Max DD (%)



Fixed Fraction - Max DD (%)



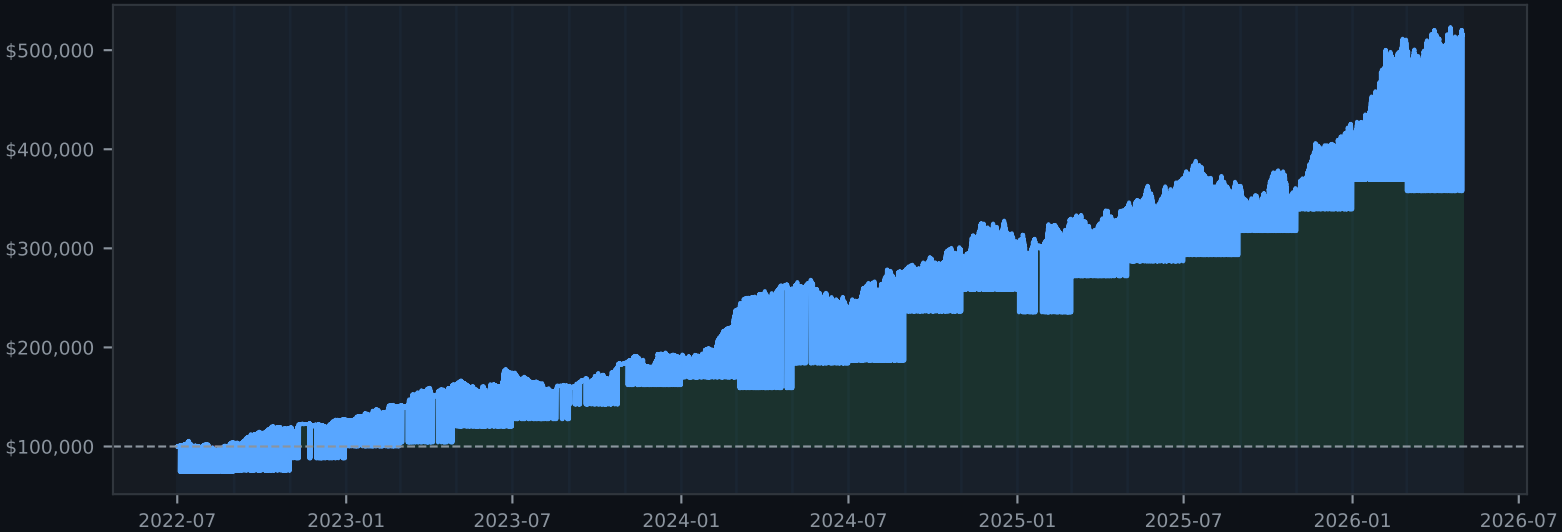
K · Best Configurations Ranking (by Sharpe Ratio)

Top 15 across all scenarios × sizing methods × leverage × risk levels

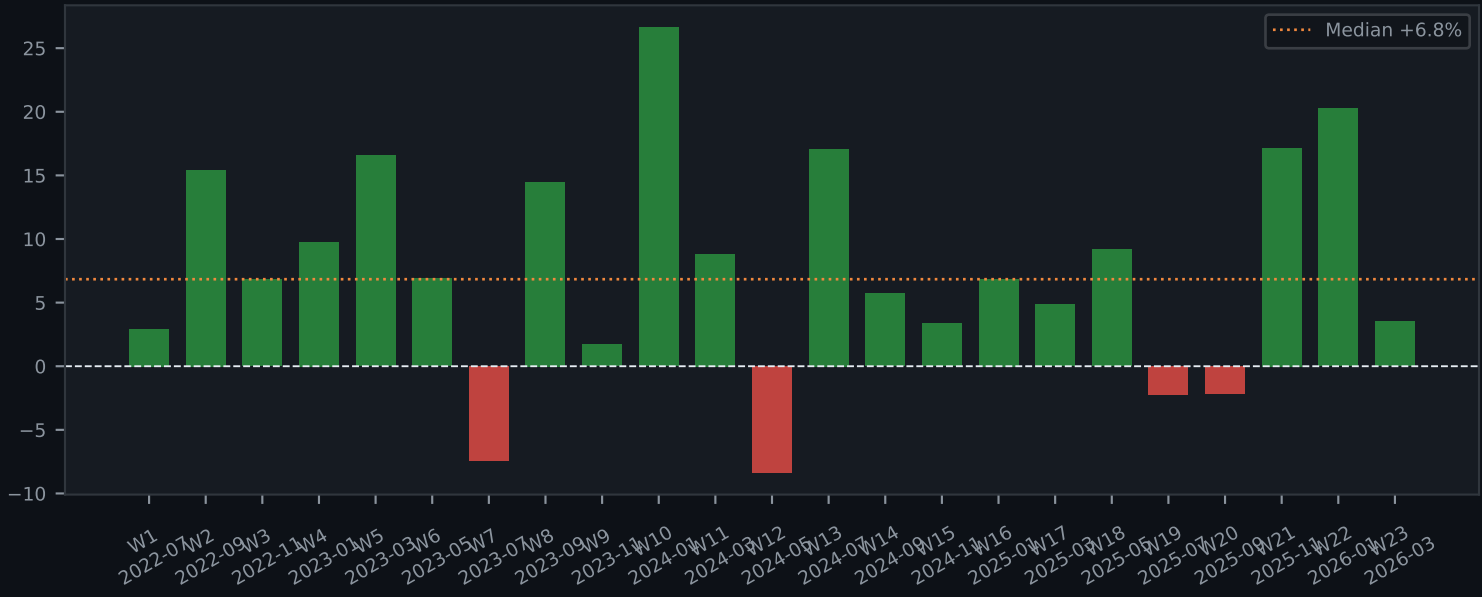


L · Walk-Forward Validation (OOS)

Chained OOS Equity — Session 08-21



OOS Return per Window (%)



23

Windows
tested

83%

Profitable
windows

+6.8%

Median OOS
return

3.835

OOS Sharpe
(combined)

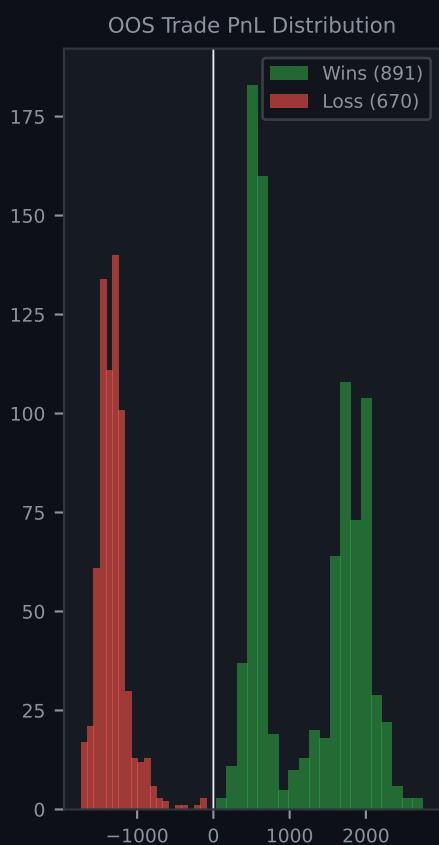
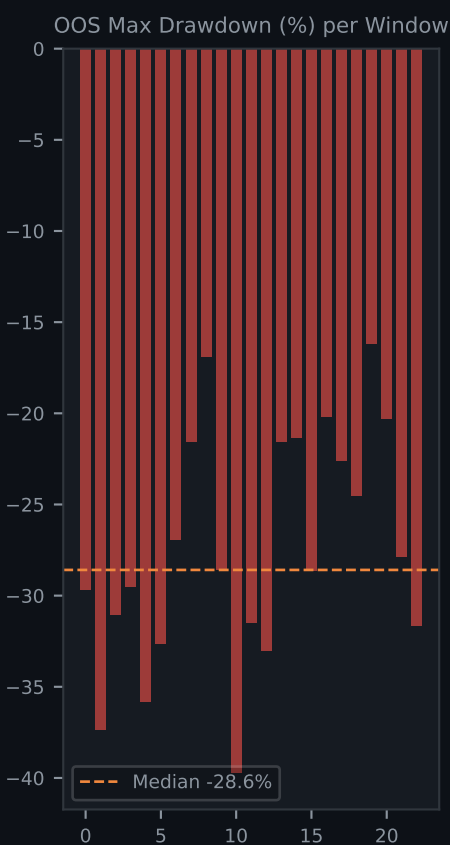
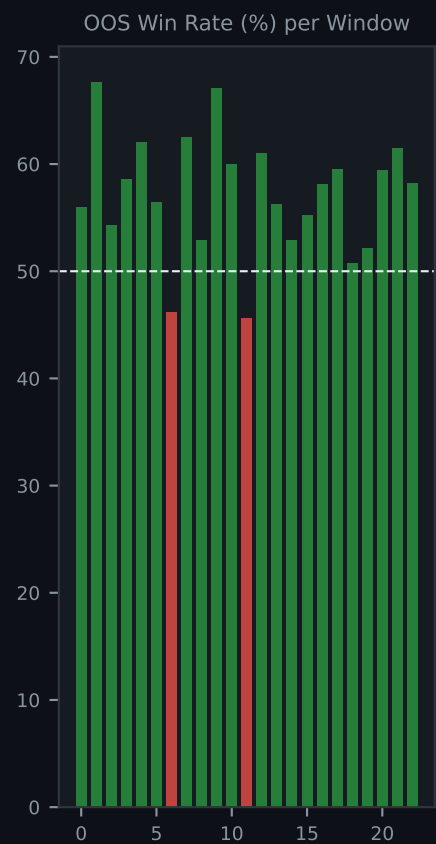
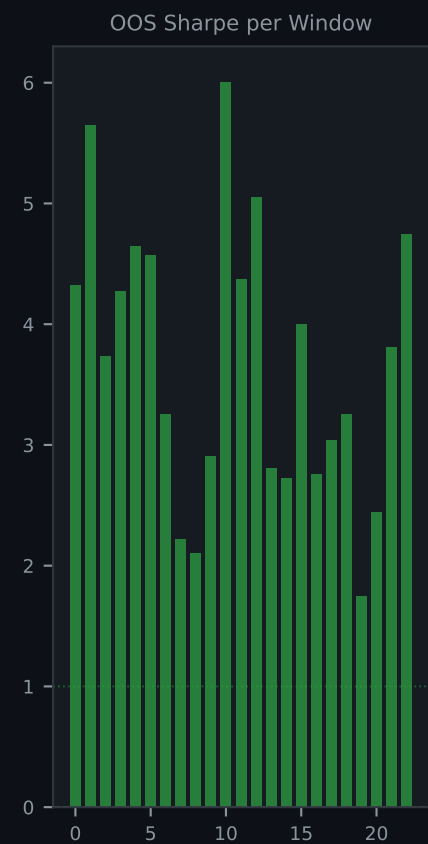
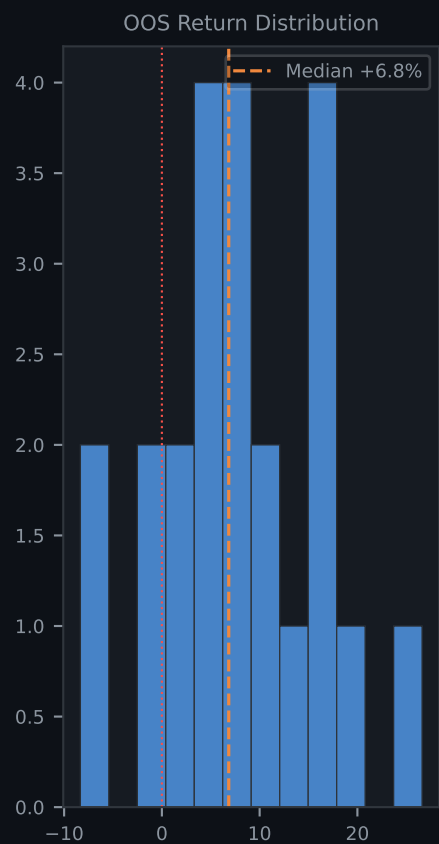
-39.7%

OOS Max DD
(combined)

+0.905

Consistency
(Sharpe of rets)

M · Walk-Forward Statistics



Per-Window KPIs					
	Ret%	N	WR%	Sharpe	DD%
W1	2.92	75.0	56.0	4.323	-29.69
W2	15.41	68.0	67.6	5.644	-37.34
W3	6.81	46.0	54.3	3.732	-31.05
W4	9.75	70.0	58.6	4.271	-29.54
W5	16.58	50.0	62.0	4.647	-35.8
W6	6.91	78.0	56.4	4.567	-32.65
W7	-7.43	65.0	46.2	3.256	-26.96
W8	14.47	48.0	62.5	2.217	-21.59
W9	1.71	70.0	52.9	2.101	-16.9
W10	26.62	70.0	67.1	2.908	-28.59
W11	8.82	60.0	60.0	6.001	-39.74
W12	-8.34	68.0	45.6	4.374	-31.49
W13	17.06	77.0	61.0	5.051	-33.02
W14	5.71	73.0	56.2	2.805	-21.55
W15	3.34	68.0	52.9	2.723	-21.34
W16	6.84	67.0	55.2	3.995	-28.66
W17	4.84	74.0	58.1	2.753	-20.21
W18	9.17	74.0	59.5	3.041	-22.63
W19	-2.22	73.0	50.7	3.254	-24.53
W20	-2.15	73.0	52.1	1.746	-16.2
W21	17.15	69.0	59.4	2.441	-20.3
W22	20.27	78.0	61.5	3.811	-27.89
W23	3.5	67.0	58.2	4.746	-31.65